

Statistical Models in Computational Biology

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Problem 15: Markov chain Monte Carlo estimation of the mean

(2 points)

Let $Y_1, Y_2, Y_3, \dots$ form a stationary Markov chain. Each Y_n has the same marginal distribution with mean $\mathbb{E}[Y_n] = \mu$ and variance $\text{Var}(Y_n) = \sigma^2$. We do not know the true mean μ in advance and want to estimate it by the Markov chain Monte Carlo estimator:

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_n.$$

(a) Prove that $\bar{Y}$ is an unbiased estimator of the mean, i.e., $\mathbb{E}[\bar{Y}] = \mu$. (0.5 points)

Next, we want to assess the variance $\text{Var}(\bar{Y})$ of our estimator given the number N of samples. If the samples $Y_1, \dots, Y_N$ were independent of each other, as in ordinary Monte Carlo estimation, $\text{Var}(\bar{Y})$ would be σ^2/N . In our *Markov chain* Monte Carlo estimation however, subsequent samples may be correlated, so the estimator will be worse with higher variance. In order to quantify this, one uses the lag- k correlations $\rho_k = \text{Corr}(Y_n, Y_{n+k})$ to define an effective sample size

$$N_{\text{eff}} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

which is typically smaller than the actual sample size N .

(b) Show that for large N we can approximate the variance of our estimator as (1.5 points)

$$\text{Var}(\bar{Y}) \approx \frac{\sigma^2}{N_{\text{eff}}}$$

under the assumption that the lag- k correlations decay at least geometrically with increasing lag k : $|\rho_k| \leq Ma^k$ for some constants $M > 0$ and $0 \leq a < 1$. From this, it follows that you may assume $\sum_{k=1}^{\infty} \rho_k$ to be finite and that for large N

$$\frac{1}{N} \sum_{k=1}^{N-1} (N-k) \rho_k \approx \sum_{k=1}^{\infty} \rho_k.$$

You can start your derivation with Bienaymé's identity:

$$\text{Var}\left(\sum_{n=1}^N Y_n\right) = \sum_{i=1}^N \sum_{j=1}^N \text{Cov}(Y_i, Y_j) = \sum_{i=1}^N \text{Var}(Y_i) + 2 \sum_{1 \leq i < j \leq N} \text{Cov}(Y_i, Y_j).$$

Problem 16: Sampling in the Rain Network

(8 points)

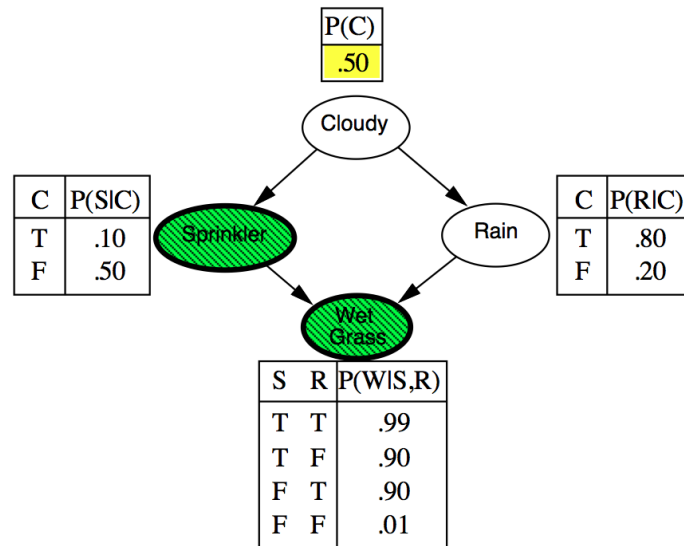


Figure 1: The *Rain Bayesian network*. Green nodes indicate that they were observed to be true.

The network shown in Figure 1 is an example of a Bayesian network that models the relationship between the variables *Cloudy* (C), *Rain* (R), *Sprinkler* (S) and *Wet grass* (W). In particular, it encodes the fact that if the grass is wet, either the sprinkler is on or it's raining, which are both influenced by whether it's cloudy or not. Each variable is then a binary variable taking values T (True) or F (False). The local probability distributions defining the Bayesian network are indicated in the figure.

Consider the task of estimating the probability of rain, given that the sprinkler is on and the grass is wet:

$$P(R = T \mid S = T, W = T).$$

In this exercise we will do this via sampling, and compare with the analytical result¹. One approach is to use Gibbs sampling to sample from $P(R, C \mid S = T, W = T)$ and marginalising over C .

We will assess whether the Gibbs sampler has reached the stationary distribution in two manners. First, we will plot the relative frequencies of $R = T$ and $C = T$ up to each iteration, for multiple independent runs of the sampler (or chains). A more formal method is the *Gelman and Rubin* multiple sequence diagnostic test [1], which compares the variance within the chains with the variance across the chains. The R package coda provides convergence diagnostics for MCMC samplers, including the *Gelman and Rubin* test.

- Derive the expressions for $P(C = T \mid R = T, S = T, W = T)$, $P(C = T \mid R = F, S = T, W = T)$, $P(R = T \mid C = T, S = T, W = T)$ and $P(R = T \mid C = F, S = T, W = T)$ and compute their values. (1 point)
- Implement the Gibbs sampler for the Bayesian network in Figure 1 and draw 100 samples from the joint posterior probability distribution $P(R, C \mid S = T, W = T)$. (1 point)

¹In general, analytical solutions are not available for interesting models.

- (c) Estimate the marginal probability of rain, given that the sprinkler is on and the grass is wet $P(R = T \mid S = T, W = T)$ from the 100 samples. (1 point)
- (d) Draw 50,000 samples instead of 100 using the Gibbs sampler.
- (e) Plot the relative frequencies of $R = T$ and $C = T$ up to each iteration t against t , for two independent runs of the sampler. Suggest a *burn-in* time based on this plot. (1 point)
- (f) Apply the *Gelman and Rubin* test and plot the potential scale reduction factor changes over the iterations using `gelman.plot()` from the `coda` package. This factor measures the ratio of the variances within and between independent runs of the sampler and should be close to 1.0 for stationary distributions. Suggest a *burn-in* time based on this plot. (1 point)
- (g) We expect adjacent members from a Gibbs sampling sequence to be positively correlated, and we can quantify this with the auto-correlation function. The lag- k auto-correlation ρ_k is the correlation between every draw and its k th neighbouring sample. Use the R function `acf()` to provide plots for both variables *Rain* and *Cloudy*. Suggest an interval for drawing approximately independent samples. (1 point)
- (h) Re-estimate $P(R = T \mid S = T, W = T)$ based on samples obtained after the suggested *burn-in* time and thinning. (1 point)
- (i) Compute the probability $P(R = T \mid S = T, W = T)$ analytically. Compare with (c) and (h) and comment on your results. (1 point)

References

- [1] Gelman, A and Rubin, DB, *Inference from iterative simulation using multiple sequences*, Statistical Science, 7, 457-511, 1992.